

Change of Variables: Jacobians

For the single integral

$$\int_a^b f(x) dx$$

you can change variables by letting $x = g(u)$, so that $dx = g'(u) du$, and obtain

$$\int_a^b f(x) dx = \int_c^d f(g(u))g'(u) du$$

where $a = g(c)$ and $b = g(d)$.

The change of variables process introduces an additional factor $g'(u)$ into the integrand.

Change of Variables: Jacobians

Jacobians:

This also occurs in the case of double integrals

$$\iint_R f(x, y) dA = \iint_S f(g(u, v), h(u, v)) \underbrace{\left| \frac{\partial x}{\partial u} \frac{\partial y}{\partial v} - \frac{\partial y}{\partial u} \frac{\partial x}{\partial v} \right|}_{\text{Jacobian}} du dv$$

where the change of variables $x = g(u, v)$ and $y = h(u, v)$ introduces a factor called the Jacobian of x and y with respect to u and v .

Change of Variables: Jacobians

Jacobians:

In defining the Jacobian, it is convenient to use the following determinant notation.

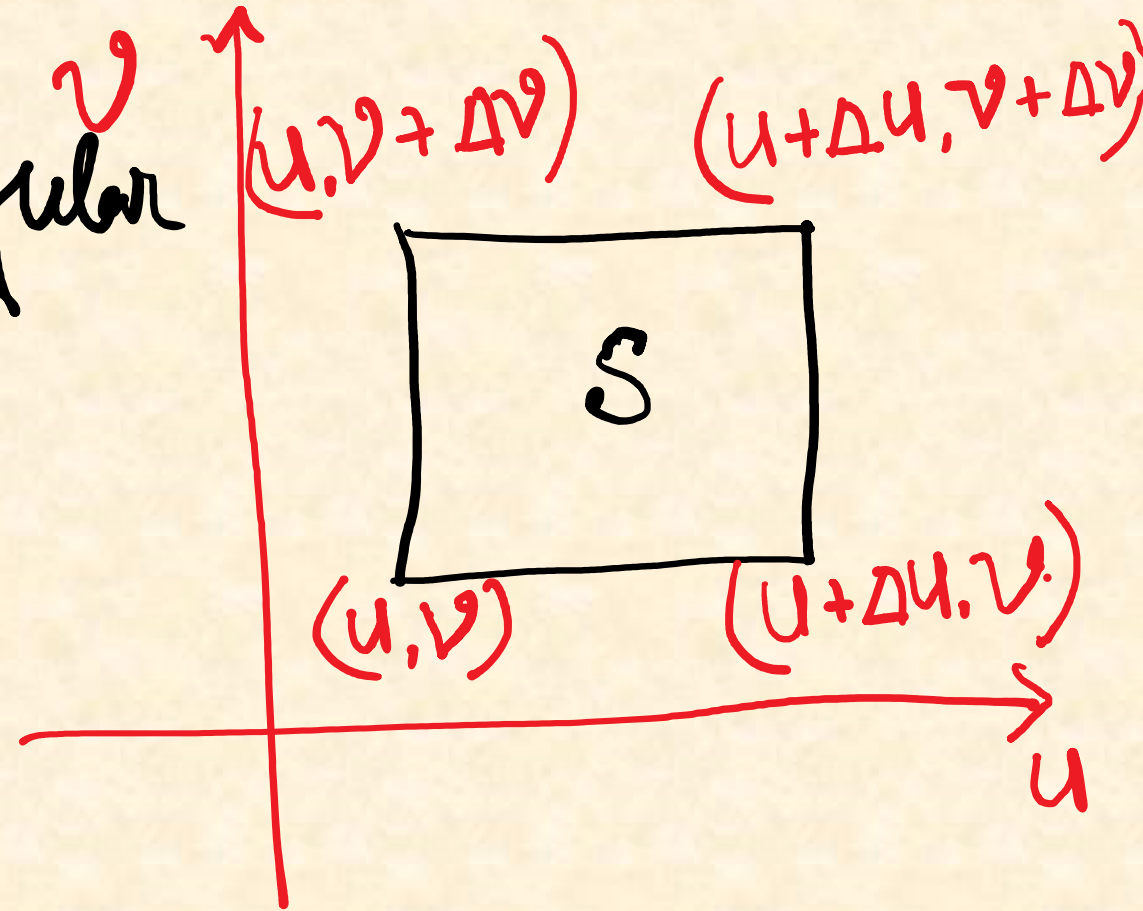
Definition of the Jacobian

If $x = g(u, v)$ and $y = h(u, v)$, then the **Jacobian** of x and y with respect to u and v , denoted by $\partial(x, y)/\partial(u, v)$, is

$$\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \frac{\partial x}{\partial u} \frac{\partial y}{\partial v} - \frac{\partial y}{\partial u} \frac{\partial x}{\partial v}.$$

Jacobians:

Proof: Let, S be a rectangular
Region in the uv -plane



Jacobians:

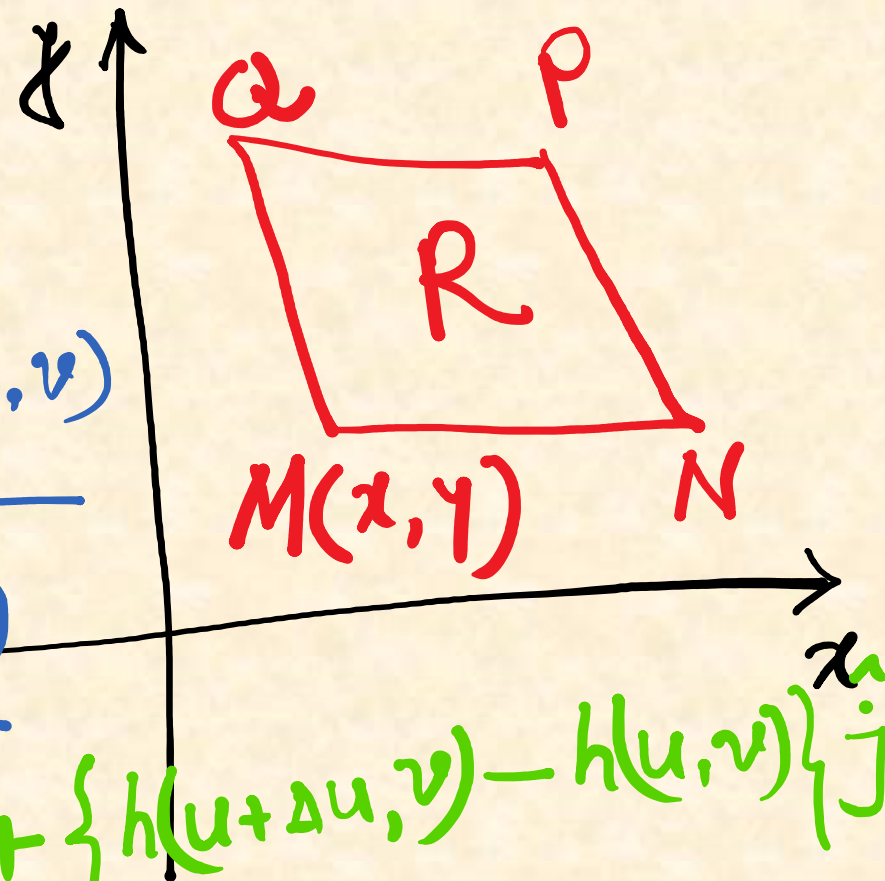
$$x = g(u, v)$$

$$y = h(u, v)$$

$$\therefore g_u(u, v) \equiv \frac{g(u + \Delta u, v) - g(u, v)}{\Delta u}$$

$$h_u(u, v) \approx \frac{h(u + \Delta u, v) - h(u, v)}{\Delta u}$$

$$\therefore \vec{MN} = \{g(u + \Delta u, v) - g(u, v)\} \hat{i} + \{h(u + \Delta u, v) - h(u, v)\} \hat{j}$$



Jacobians:

$$\approx [g_u(u,v) \Delta u] \hat{i} + [h_u(u,v) \Delta u] \hat{j}$$

$$= \left[\frac{\partial x}{\partial u} \Delta u \right] \hat{i} + \left[\frac{\partial y}{\partial u} \Delta u \right] \hat{j}$$

Similarly, $\vec{MQ} \approx \left[\frac{\partial x}{\partial v} \Delta v \right] \hat{i} + \left[\frac{\partial y}{\partial v} \Delta v \right] \hat{j}$

$$\therefore \Delta A \approx |\vec{MN} \times \vec{MQ}|$$

Jacobians:

$$\therefore \vec{MN} \times \vec{MQ} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial x}{\partial u} \Delta u & \frac{\partial y}{\partial u} \Delta u & 0 \\ \frac{\partial x}{\partial v} \Delta v & \frac{\partial y}{\partial v} \Delta v & 0 \end{vmatrix} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} \end{vmatrix} \begin{vmatrix} \Delta u \Delta v & \hat{k} \end{vmatrix}$$

$$\therefore dA \approx |\vec{MN} \times \vec{MQ}| \approx \left| \frac{\partial(x,y)}{\partial(u,v)} \right| du dv$$

$$\therefore \iint_R f(x,y) dx dy = \iint_S f(g(u,v), h(u,v)) \left| \frac{\partial(x,y)}{\partial(u,v)} \right| du dv$$

Jacobians:

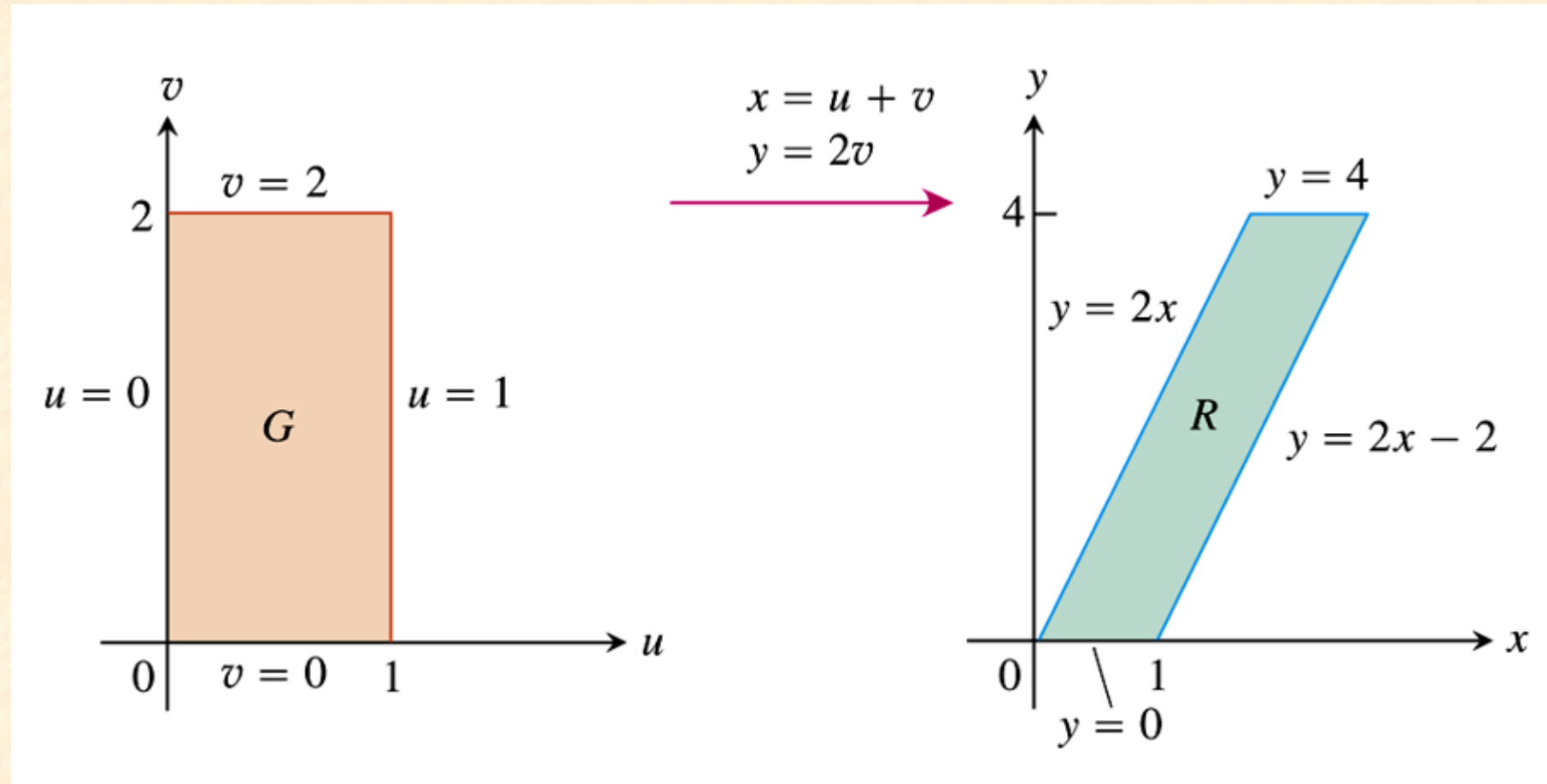
Similarly for Triple integral

$$\iiint_D F(x, y, z) dx dy dz = \iiint H(u, v, w) |J(u, v, w)| du dv dw$$

$$\text{where } J(u, v, w) \approx \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} & \frac{\partial z}{\partial u} \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} & \frac{\partial z}{\partial v} \\ \frac{\partial x}{\partial w} & \frac{\partial y}{\partial w} & \frac{\partial z}{\partial w} \end{vmatrix} = \frac{\partial(x, y, z)}{\partial(u, v, w)}$$

Jacobians:

Example 1: Evaluate $\int_0^4 \int_{x=y/2}^{x=(y/2)+1} \frac{2x-y}{2} dx dy$ by applying the Jacobian transformation.



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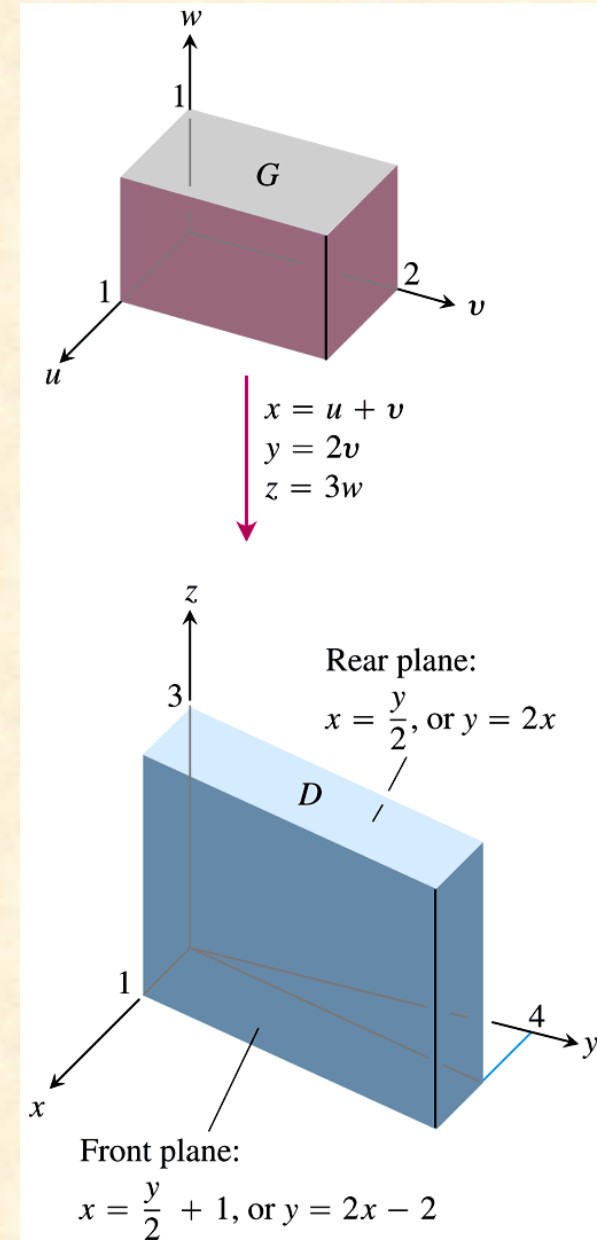
Jacobians:

Jacobians:

Jacobians:

Example 2: Evaluate $\int_0^3 \int_0^4 \int_{x=y/2}^{x=(y/2)+1} \left(\frac{2x-y}{2} + \frac{z}{2} \right) dx dy dz$

by applying the Jacobian transformation and integrating over an appropriate region in uvw -space.



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Thanks a lot ...